

Detailed Notes Based on Uploaded Images

1. Harmful Prompts and AI Misuse

Definition

Harmful prompts are instructions given to an AI system that encourage unsafe, unethical, misleading, or malicious behavior. AI misuse occurs when AI tools are intentionally or unintentionally used to create harmful outcomes. **Major Risks**

- **Misinformation and Bias:** AI can generate fake news, distorted facts, or biased conclusions when trained on biased data.
- **Data Misuse and Attacks:** Attackers may use AI for phishing, malware generation, social engineering, or unauthorized data extraction.
- **Harmful Content Creation:** AI can produce offensive, deceptive, manipulative, or dangerous content if safeguards are bypassed.

Prompt Monitoring

Organizations implement monitoring systems to detect suspicious prompts, policy violations, and unsafe requests before generating responses. **Fact Verification**

AI-generated information should always be verified using trusted sources because language models can hallucinate or produce inaccurate outputs. **Responsible AI Practices**

- Follow ethical guidelines.
- Respect privacy and security.
- Avoid generating harmful instructions.
- Maintain transparency and accountability.
- Keep humans involved in important decisions.

Key Takeaway

AI systems should be used responsibly, with safeguards that reduce misinformation, attacks, harmful outputs, and misuse.

2. One-Shot Prompting

Definition

One-shot prompting is a prompting technique where a model is given exactly one example before it is asked to perform a task. **Purpose**

The example teaches the model the desired format, style, or reasoning pattern for the expected answer. **How It Works**

Example Pair:

Input: "What is the capital of France?"

Output: "Paris"

Task Pair:

Input: "What is the capital of Japan?"

Expected Output: "Tokyo" **Advantages**

- Improves response accuracy.
- Clarifies expected output format.
- Reduces ambiguity in instructions.
- Useful when the task structure is important. **Comparison**
- Zero-shot prompting: No examples are provided.
- One-shot prompting: One example is provided.
- Few-shot prompting: Multiple examples are provided. **Applications**

Classification, translation, question answering, information extraction, summarization, and structured output generation.

3. Transformer Workflow Visualization

Transformer Architecture Workflow

1. Input Text

The process begins with raw text entered by the user.

2. Tokenization

The text is divided into smaller units called tokens (words, subwords, or symbols).

3. Embedding

Each token is converted into a numerical vector representation that captures semantic meaning.

4. Self-Attention Mechanism

The model learns relationships between tokens and determines which words are most important for understanding context.

5. Feed Forward Layers

The transformed information passes through neural network layers that refine feature representations.

6. Output Prediction

The model predicts the next token, classification label, translation, or other task-specific output.

Why Self-Attention Matters

Self-attention allows the model to capture long-range dependencies and contextual relationships more effectively than traditional sequence models.

Practical Significance

Transformers power modern AI systems such as large language models, machine translation systems, summarizers, chatbots, and code assistants.

Workflow Summary

Input Text → Tokenization → Embedding → Self-Attention → Feed Forward Processing → Output Prediction